



Intersubband polariton-LO phonon interaction in mid-infrared non-dispersive cavities: experimental demonstration of spontaneous scattering and perspectives towards polariton lasing [Invited]

PAUL GOULAIN,¹ MATHIEU JEANNIN,¹  STEFANO PIROTTA,¹ ADEL BOUSSEKSOU,¹ GIORGIO BIASIOL,² IACOPO CARUSOTTO,³  RAFFAELE COLOMBELLI,¹  AND JEAN-MICHEL MANCEAU^{1,*} 

¹Centre de Nanosciences et de Nanotechnologies, CNRS UMR 9001, Université Paris Saclay, 10 Boulevard Thomas Gobert, 91120 Palaiseau, France

²Laboratorio TASC, CNR-IOM, Area Science Park, S.S. 14 km 163.5, Basovizza I-34149 Trieste, Italy

³Pitaevskii BEC Center, CNR-INO and Dipartimento di Fisica, Università di Trento, I-38123 Trento, Italy

*jean-michel.manceau@cnrs.fr

Abstract: We report experimental evidence of the interaction between intersubband polaritons and longitudinal optical phonons in non-dispersive mid-infrared cavities, under resonant optical injection. Light emission originating from spontaneous polariton-phonon scattering is observed at a frequency corresponding to an energy shift of one phonon below the pump frequency. Given the extremely low spontaneous scattering rate, we employ a custom-developed quantum mechanical model to numerically demonstrate the feasibility of stimulating such a process using a pump-probe scheme. Based on this theoretical framework, we identify a set of experimental conditions under which optical gain may be realized in an intersubband polaritonic system.

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1. Introduction

Over the past decades, the precise control of light-matter interactions has evolved into a central theme in photonics. In the weak coupling regime, where the interaction strength between an emitter and a confined photonic mode remains below the dissipative rates, quantum electrodynamical effects such as the Purcell enhancement govern the modification of spontaneous emission rates. This regime enables efficient control over radiative processes and has been widely exploited in systems ranging from atomic cavities to solid-state quantum emitters [1,2]. As the coupling strength increases and surpasses the combined loss rates of the emitter and the cavity, the system enters the strong coupling regime, characterized by the reversible and coherent exchange of excitations between light and matter. This regime is marked by the emergence of hybridized excitation eigenmodes — polaritons — exhibiting normal-mode splitting in the spectral domain, with the Rabi frequency setting the energy scale of the interaction. Strong coupling has been observed across several platforms [3–5], and in particular with solid-state systems offering versatility thanks to their compatibility with advanced nanofabrication. Progress in lithographic patterning and epitaxial growth has enabled the fine-tuning of both photonic resonators and quantum emitters, facilitating systematic access to this non-perturbative regime [6].

In the mid- and far-infrared (MIR–FIR) spectral ranges, intersubband (ISB) transitions in semiconductor quantum wells (QWs) constitute a particularly interesting platform for exploring strong coupling phenomena, owing to their large oscillator strengths and tunable transition energies. This system, which forms the backbone of quantum cascade lasers (QCL) [7], has proven especially useful for exploring quantum electrodynamical phenomena [8,9]. A key feature

of ISB polaritons is that the Rabi frequency is directly proportional to the population difference Δn between the two subbands:

$$\Omega_{\text{Rabi}}^2 = f_w \frac{f_{12} \Delta n e^2}{4\epsilon\epsilon_0 m^* L_{\text{QW}}} \quad (1)$$

where f_w is the ratio of the total quantum well length to the overall stack thickness, f_{12} is the transition oscillator strength, L_{QW} is the length of a single quantum well, and m^* is the effective electron mass in the semiconductor under consideration [10]. The population difference Δn originates from the charge density introduced by the donor dopants: these are located in either the well or the barrier and establish a two-dimensional electron gas within the well. This was early on identified as a tuning parameter with significant implications both from a fundamental and applied perspective. In practice, applying a bias allows the QWs to be depleted, enabling a variety of experiments where the Rabi frequency is electrically modulated. This modulation permits the dynamic control of the system, toggling it in and out of the strong coupling regime. Such control proves particularly useful for implementing amplitude or phase modulation on a continuous-wave (CW) carrier at GHz frequencies [11–14]. More recently, the same concept of entering and exiting the strong coupling regime has been proposed as a means to implement ultrafast saturable absorber mirrors, this time by optically saturating the transition [15–20].

ISB polaritons have also been the subject of more fundamental investigations, especially concerning their bosonic nature. A pioneering theoretical study of the quantum properties of such systems [21] identified the possibility to increase the coupling strength way above the cavity and emitter decoherence rates up to the bare cavity and emitter frequencies and beyond [22]. In this regime, the effects of the counter-rotating terms in the Hamiltonian—typically neglected—become significant and potentially observable. This so-called ultra-strong coupling regime holds the promise of extending quantum effects and technologies into the FIR range of the spectrum, offering possibilities such as the emission of correlated photon pairs through abrupt modulation of the system, or the electrical probing of quantum phase transitions [23]. Demonstrating this regime has been the focus of significant efforts by the ISB community over the past decade [24–26], with successful implementations across a variety of microcavity configurations and material systems [27–31]. Equally interesting are theoretical proposals leveraging the bosonic nature of these systems to reach final-state stimulation and inversionless lasing [32]. Initial attempts to realize such devices through electrical carrier injection have thus far only yielded electroluminescence with relatively low efficiency [33–35]. More recent approaches focus on resonant optical pumping, exploiting interactions with longitudinal optical phonons or through polariton-polariton interaction processes [36,37].

In the present work, we demonstrate light emission from spontaneous polariton-phonon scattering in a non-dispersive microcavity—an experimental approach that has remained unexplored until now. By directly measuring the light emitted by polaritons scattered through their coupling with LO phonons, we observe a distinct frequency locking of the emission exactly one LO-phonon below the pump frequency. Owing to the flat, non-dispersive nature of the photonic mode, this emission remains fixed in frequency even when varying the pump incidence angle—an effect with no equivalent in conventional dispersive systems. Beyond the experimental evidence, we extend the generalized Optical Bloch Equations framework developed in Ref. [37] to include the ISB-phonon interaction, enabling a numerical exploration of the phonon-assisted scattering dynamics. Our simulations reveal that, under appropriate excitation conditions, stimulated scattering and net optical gain can be achieved. Together, these results establish a new platform for phonon-assisted polaritonic amplification in flat-band photonic systems.

2. Design, fabrication, and characterization of the strongly coupled system

2.1. Multiple quantum wells stack

To conduct this study, a quantum well stack was epitaxially grown. It consists of a multiple QW system: 7 repetitions of 8.3-nm-wide GaAs QWs separated by 20-nm-thick $\text{Al}_{0.33}\text{Ga}_{0.67}\text{As}$ barriers. The nominal surface doping per QW ($n_{2D} = 4 \times 10^{12} \text{ cm}^{-2}$) is introduced as δ -doping layers placed at the center of the barriers. The total thickness of the active region is 288 nm. A numerical simulation using a commercial Poisson–Schrödinger solver (NextNano software) is shown in Fig. 1(a). The simulation includes the Hartree potential correction arising from ionized impurities in the barriers. The finite electron density in the two-dimensional electron gas then leads to a renormalization of the transition frequency, commonly referred to as the depolarization shift [10,38,39], given by the expression: $\tilde{\omega} = \sqrt{\omega_{21}^2 + \omega_p^2}$, where ω_p is the plasma frequency of the 2D electron gas. Overall, the ISB transition frequency $\tilde{\omega}$ is expected to occur around 27 THz.

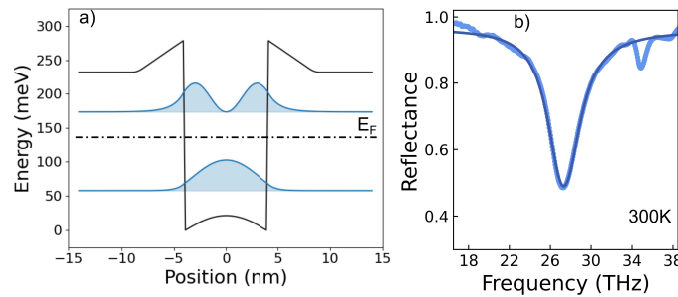


Fig. 1. (a) Simulated energy band profile and wavefunctions. (b) Transmittance of the sample shaped in a multipass prism configuration.

To evaluate the optical performance of the fabricated structures, the samples were shaped into a multi-pass prism, with facets polished at a 45° angle. A titanium/gold metallization was deposited on the top surface to enhance the overlap of the electric field with the QW stack. A polyethylene polarizer was used to select the polarization of the Globar source. The interferogram were recorded using a Fourier Transform Infrared (FTIR) spectrometer.

Due to the selection rules of intersubband transitions, the QWs selectively absorb transverse magnetic (TM) polarized light. To isolate this absorption, transmission spectra for TE- and TM-polarized incident light were measured, and their ratio was calculated. Figure 1(b) shows the resulting transmittance spectrum at room temperature, with a peak centered at 27.3 THz, in good agreement with the numerical simulation. From a Voigt profile fit, the linewidth of the transition is estimated to be 3 THz, corresponding to 10.9% of the central frequency. Note that the feature observed around 35 THz is a spurious absorption line coming from the sample mount.

2.2. Polaritons formation in the resonant optical micro-cavity

Metal-Insulator-Metal (MIM) cavities are a class of photonic structures widely used in conjunction with MIR and FIR QW-based devices due to their ability to confine electromagnetic modes with a non-zero electric field component in the z -direction—crucial for interacting with the ISB transition. In these structures, waveguiding is achieved through boundary conditions imposed by the metallic layers, which restrict the propagation direction of light. The bottom mirror in these cavities is typically a flat gold layer while the optical access to the microcavity is made possible by periodically patterning the top mirror along one direction (from now on called x). Depending on its thickness, the microcavity can operate in either a dispersive or non-dispersive regime. In the dispersive configuration, the photonic dispersion can be fully explored as a function of the

incident angle [40]. In contrast, in the non-dispersive regime, the cavity remains resonant at a fixed frequency regardless of the angle of incoming light along the xz plane.

The latter is the case of interest for this work. As shown on Fig. 2(a), the resonant mode (E_z) is strictly confined below each metallic stripe, which will act as an independent cavity as extensively discussed in [41]. In the case of a 1D stripe with infinite length along y , the mode resonance frequencies depend on the stripe width and follow the formula:

$$\nu_M = \frac{Mc}{2n_M\Lambda} \quad (2)$$

where M is the mode index, n_M the effective refractive index of the mode, and Λ the width of the stripe. Details of the fabrication process can be found in [42].

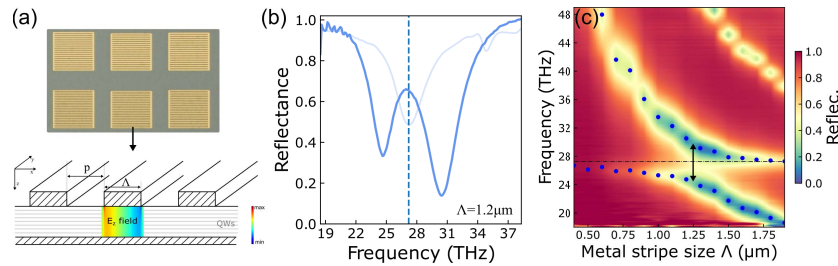


Fig. 2. (a) Optical image showing 6 microcavity arrays (view from top), each having a different metal stripe size. Below is a schematic side view of the microcavity system. (b) Reflectance measurement of the microcavity device with a stripe size of $1.2 \mu\text{m}$, along with the transmittance of the bare ISB transition (in transparency). The dash line marks its central frequency. (c) Reconstructed experimental reflectance as function of the strip size. The arrow marks the position of the anti-crossing point.

In order to record the characteristic anticrossing behavior of polaritons, an array of microcavities with increasing stripe widths was fabricated, while maintaining a constant spacing p of $1.4 \mu\text{m}$ between the stripes. A photograph of the sample is shown in Fig. 2(a), where six different microcavities can be seen. A side view schematic of the microcavity is shown in the lower panel. Figure 2(b) presents the reflectance spectrum of the microcavity with a stripe width of $1.2 \mu\text{m}$, measured using a FTIR spectrometer equipped with a microscope. The reflectance was obtained by normalizing the sample's reflectivity to that of a flat gold mirror. A clear lifting of degeneracy is observed, with the appearance of new mixed eigenstates. By collecting together the measurements done on each microcavity device, the polaritonic dispersion as a function of the cavity size can be reconstructed, revealing the classic anticrossing behavior of the hybrid states as shown on Fig. 2(c). From the minimum energy splitting, we infer the Rabi frequency of the system to be 2.65 THz .

3. ISB polariton-LO phonon scattering under resonant optical pumping condition

In order to study the spontaneous phonon scattering processes under resonant light injection, we built an optical setup, presented in Fig. 3(a). The laser source is a commercial QCL from Daylight Solutions, tunable from 31 to 34.5 THz , with a maximum average output power in pulsed mode (100 kHz , $2 \mu\text{s}$) of 110 mW . A compact goniometric arm directs the QCL output to precise positions in the energy-momentum space of the polaritonic system. The specular component of the reflected signal is measured using a power meter, enabling accurate alignment with the resonant polaritonic mode. Using a beam profiler, we measured the beam waist at the focal point to be $170 \mu\text{m}$ ($1/e^2$), when focused with ZnSe lens ($f/\#2$). The resulting average

pump intensity can reach up to 750 W/cm^2 (3.75 kW/cm^2 peak power). The light emitted by the scattered polaritons is collected at normal incidence using a ZnSe lens ($f/\#1.5$) and analyzed with a FTIR spectrometer coupled to a mercury cadmium telluride detector cooled to 78 K. A Long-wave pass filter is employed to reduce the amount of stray light scattered from the sample surface that reaches the detector.

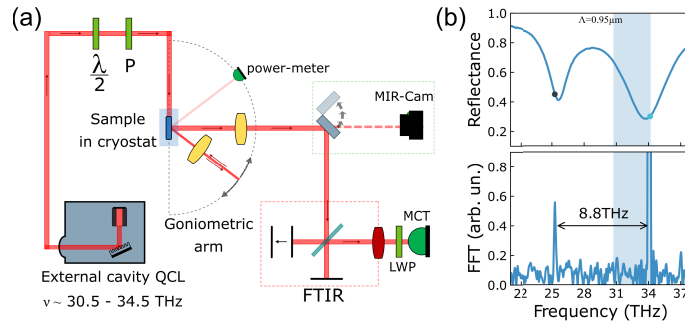


Fig. 3. (a) Experimental optical set-up with the tunable QCL and goniometric arms that allow the different experimental pump positions. (b) Upper panel, experimental reflectance of the microcavity with $\Lambda = 0.95 \mu\text{m}$ lateral size, recorded at 78K. The light blue dot corresponds to the pump state while the dark blue one is the final state. Lower panel, FFT of the recorded interferogram of the sample under resonant injection of light at 34.2 THz. The main peak is the remaining stray light from the pump. The lower peak corresponds to the photons emitted by the small fraction of polaritons scattered via LO-phonons. The blue shaded area is the QCL tuning range.

Our first experiment aimed to confirm the possibility of detecting light emission from polaritons that are scattered into a final state after emitting a longitudinal optical phonon (LO_{ph}). The upper panel of Fig. 3(b) displays the reflectance spectrum of the polaritonic system in a microcavity with a stripe width of $0.95 \mu\text{m}$. This particular microcavity was selected because it provides an energy separation between the pumped and the final states of one LO phonon energy, while remaining within the tuning range of the QCL. The tuning range of the pump is indicated by the shaded blue area, and the positions of both the pump and the final states are marked with dots. Figure 3(b) (lower panel) shows the emission spectrum collected at normal incidence. The main peak at 34.2 THz corresponds to residual stray light from the pump using only one long pass filter. The lower-frequency peak originates from light emitted by the small fraction of polaritons that have scattered via LO_{ph} . The peak separation of 8.75 THz matches the LO_{ph} energy in GaAs.

We further investigated this polaritonic configuration by tuning both the frequency and the in-plane momentum of the injected light. Figure 4(a) shows the numerically calculated polaritonic band-structure of the sample and the different positions in frequency and angle that have been investigated. The left panel of Fig. 4(b) shows the emitted light from the scattered polaritons remains strictly locked to the LO_{ph} energy separation when varying the pump frequency. More notably, when tuning the in-plane momentum of the injected polaritons along x , the signal remains locked to the LO_{ph} energy, highlighting the non-dispersive nature of the cavity mode (Figure 4b right panel). One potential problem that arises from such non-dispersive devices is that the flat dispersion implies a continuum of possible final states at different k_x values. This continuum of almost-degenerate final states implies that there is no preferential state for polaritons to scatter to. Hence, scattered polaritons are spread across a large part of the angular range, and thus the spontaneous emission is also spread across a wide range of angles.

To test this hypothesis, a measurement was conducted using the same pumping angle, frequency, and power, but with a different collection lens: a 1-inch ZnSe lens instead of the previously used 2-inch one (while preserving the focal length), hence reducing the cone of collection of light

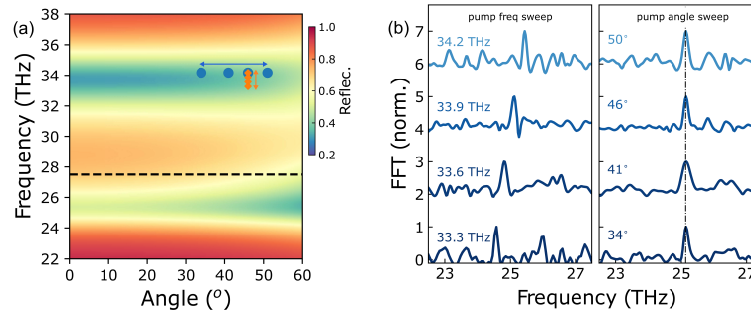


Fig. 4. (a) Simulated reflectance as a function of the incidence angle in the xz plane. The different experimental pump positions are indicated. (b) Left panel, emission spectra when the pump frequency is changed while the angle is kept constant at 41° . Right panel, emission spectra when the pump angle is changed while frequency is kept constant at 33.9 THz. The spontaneous emission is locked 8.75 THz below the pump.

by a factor 2. In such configuration, no signal could be detected. Furthermore, although our initial hypothesis was that flattening the dispersion of the final state might enhance the scattering efficiency, no significant improvement was observed. The conversion efficiency of the process remains on the order of 10^{-8} , consistent with values reported for dispersive cavities [36]. This outcome underscores the importance of enabling the selection of a specific final state for polariton scattering, which could be achieved by adopting an experimental approach, where a probe pulse would populate the final state of choice.

4. Achieving gain conditions with a pulsed pump-probe configuration

In stark contrast to fermions, bosons exhibit stimulated scattering, wherein the transition probability from an initial to a final state increases with the occupation of the final state. Specifically, if a scattering channel is available from a reservoir to a lower-energy state with a spontaneous rate Γ_0 , the effective scattering rate can be significantly enhanced by the population density n_f already present in the final state. This bosonic stimulation mechanism has been extensively harnessed in the exciton-polariton community, leading to the realization of non-equilibrium Bose-Einstein condensation [43–48], opening the way to a variety of quantum many-body physics effects [49] and offering an interesting platform for photonic quantum simulation [50].

As thoroughly described in [21], ISB polaritons can be approximated as bosons in the dilute regime, making them subject to bosonic final-state stimulation and potentially enabling inversionless lasing [32,51]. While several demonstrations of electroluminescent devices have been reported, no threshold behavior indicative of stimulated coherent emission has been observed [33,34,36]. This is primarily because all devices developed to date have relied on resonant electrical injection into optically bright polaritonic states—a process that is extremely challenging. Even the use of narrow electrical injectors provides little improvement, as the majority of the electrical energy is still injected into the numerous electronic dark states of the system [52]. This often results in thermally assisted electroluminescent devices [35]. A more promising scheme involve direct polariton-polariton scattering [37,39,53,54] or the polariton-phonon scattering process [32,36] under resonant optical pumping scheme.

In light of the parallels with exciton-polariton systems [55], a critical requirement for demonstrating stimulated scattering is the ability to populate the final state through a synchronized pump-probe experimental approach. In the case of ISB polaritons, such a regime was recently achieved using a non-collinear pump-probe geometry with phase-stable mid-IR pulses [37]. The

experimental observations were quantitatively recovered by a quantum-mechanical model of polariton-polariton scattering based on the Optical Bloch Equation (OBE) formalism. In this work, we extend this theoretical model to include the LO phonon-polariton interaction and we numerically investigate the set of experimental conditions under which gain could be observed.

4.1. Theoretical formalism

In this subsection, we start by introducing a general theoretical framework describing the coupling of intersubband polaritons to phonons. The readers that are not interested in the technical details can focus on the main results, namely the spatially local form of the phonon-polariton coupling Hamiltonian (21) and the expression (20) for the coupling constant.

4.1.1. Frölich coupling between phonons and intersubband excitations

Our theoretical formalism is based on the Frölich coupling between electrons and phonons, which in a three-dimensional geometry has the standard form [56,57]

$$\begin{aligned} H_{e,ph} &= \int d^3r \int \frac{d^3q}{(2\pi)^3} \alpha(q) e^{-iqr} b_q^\dagger n_e(r) + \text{H.c.} = \\ &= \int \frac{d^3q}{(2\pi)^3} \alpha(q) \int \frac{d^3k}{(2\pi)^3} b_q^\dagger a_k^\dagger a_{k+q} + \text{H.c.} \end{aligned} \quad (3)$$

with a coupling constant

$$\alpha(q) = \left[\frac{2\pi\hbar\omega_{phon}e^2}{\epsilon_p q^2} \right]^{1/2}. \quad (4)$$

Here, ω_{phon} is the (q -independent) longitudinal-optical (LO) phonon frequency, the phononic contribution to the dielectric constant is written as $\epsilon_p^{-1} = [\epsilon_\infty^{-1} - \epsilon_s^{-1}]$ in terms of the static (ϵ_s) and high-frequency (ϵ_∞) dielectric constants, and the electron density is

$$n_e(r) = \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3} e^{i(k'-k)r} a_k^\dagger a_{k'} \quad (5)$$

In order to develop a two-dimensional formalism, we can express electron operators on the subband basis along z ,

$$a_{K,k_z} = \sum_j \tilde{\phi}_j(k_z) a_{K,j} \quad (6)$$

where j runs over the subbands of the quantum well potential of wavefunction $\phi_j(z)$ and

$$\tilde{\phi}_j(k_z) = \int dz e^{-ik_z z} \phi_j(z) \quad (7)$$

is the wavefunction of the j subband expressed in reciprocal space.

Inserting this expansion into the Frölich Hamiltonian (3), we obtain

$$H_{e,ph}^{2D} = \int \frac{d^2Q}{(2\pi)^2} \sum_{j,j'} C_{j,j'}(Q) \beta_{Q,j,j'}^\dagger \int \frac{d^2K}{(2\pi)^2} a_{K,j}^\dagger a_{K+Q,j'} + \text{H.c.} \quad (8)$$

where for each j, j' pair and a given Q the phonon creation operators are defined as suitable superpositions of q_z modes matching the electronic transition,

$$\beta_{Q,j,j'}^\dagger = \frac{1}{C_{j,j'}(Q)} \int \frac{dq_z}{(2\pi)} I_{j,j'}(q_z) \alpha(Q, q_z) b^\dagger(Q, q_z). \quad (9)$$

The coupling coefficient $C_{j,j'}(Q)$ is determined by imposing Bose commutation rules

$$[\beta_{Q,j,j'}, \beta_{Q',j,j'}^\dagger] = (2\pi)^2 \delta^{(2)}(Q - Q') \quad (10)$$

for the phononic operators, which gives

$$\begin{aligned} C_{j,j'}(Q) &= \left[\int \frac{dq_z}{(2\pi)} |\mathcal{I}_{j,j'}(q_z) \alpha(q_z, Q)|^2 \right]^{1/2} = \\ &= \left[\frac{\pi \hbar \omega_{phon}^2}{e} Q \epsilon_\rho \int dz \int dz' e^{-Q|z-z'|} \phi_j^*(z) \phi_{j'}(z) \phi_{j'}^*(z') \phi_j(z') \right]^{1/2} \end{aligned} \quad (11)$$

in terms of the overlap factors

$$\mathcal{I}_{j,j'}(q_z) = \int \frac{dk_z}{(2\pi)} \tilde{\phi}_j^*(k_z) \tilde{\phi}_{j'}(k_z + q_z) = \int dz \phi_j^*(z) \phi_{j'}(z) e^{-iq_z z}. \quad (12)$$

The Hamiltonian (8) has the simple physical interpretation of an electron scattering from the j' to the j subband while emitting a phonon of in-plane momentum Q (and viceversa). For each Q , the wavefunction along k_z of the selected phonon mode is determined by the coupling to the electronic transition via (9). As a consequence of the momentum-independence of the LO phonon frequency ω_{phon} , phonons do not propagate away from the QW.

In this work we are interested in phonon scattering processes involving intersubband polaritons that originate from the coupling of light to collective intersubband transitions. In terms of electronic operators, the creation operator of a ISB excitation from the first (1) to the second (2) subband has the form [10]:

$$B_k^\dagger = \frac{1}{\sqrt{n_{el}}} \int_{FS} \frac{d^2 K}{(2\pi)^2} a_{K+k,2}^\dagger a_{K,1} \quad (13)$$

where integration over K is restricted to $|K| < k_F$ within the Fermi sphere and the sum over electron spin states is kept implicit for notational simplicity. For simplicity, we restrict to a low-electron-density regime where the Coulomb interactions do not significantly distort the subband wavefunctions. In the weak excitation limit where the density of ISB excitations on top of the filled Fermi sea in the first subband is much smaller than the electron density n_{el} , the ISB operators satisfy Bosonic commutation rules

$$[B_k, B_{k'}^\dagger] = (2\pi)^2 \delta^{(2)}(k - k'). \quad (14)$$

Physically, an ISB excitation at k corresponds to a symmetric superposition of all possible electrons being promoted from K to $K + k$. Given the K -independence of the coupling amplitude in the Fröhlich Hamiltonian (8), interaction with the phonons does not couple ISB excitations to the (non-symmetric) dark states.

In contrast to recent works focused on the (ultra)strong-coupling effects in the coherent interconversion between ISB polaritons and phonons and on the Rabi splitting of the resulting ISB polaron modes [58,59], we are interested here in processes where ISB polaritons scatter via emitting/absorbing a phonon but the number of polaritons is not changed. Having such processes in mind and assuming that the frequency of ISB excitations is well above the LO phonon frequency, we can restrict the sum in (8) to the diagonal $j = j' = 1$ and $j = j' = 2$ terms while neglecting the non-diagonal ones underlying the ISB plasmon-polaritons [58,59] which are in our case off-resonance.

Within this approximation, the electronic term of (8) can be expressed in terms of ISB operators as

$$H_{e,ph}^{2D} = \int \frac{d^2Q}{(2\pi)^2} \left[C_{2,2}(Q) \beta_{Q,2,2}^\dagger - C_{1,1}(Q) \beta_{Q,1,1}^\dagger \right] \int \frac{d^2K}{(2\pi)^2} B_K^\dagger B_{K+Q} + \text{H.c.} \quad (15)$$

where the two terms correspond to scattering of an electron in the second subband or of a hole in the first subband. This Hamiltonian can be further simplified into

$$H_{e,ph}^{2D,pol} = \int \frac{d^2Q}{(2\pi)^2} \tilde{C}(Q) \tilde{\beta}_Q^\dagger \int \frac{d^2K}{(2\pi)^2} B_K^\dagger B_{K+Q} + \text{H.c.} \quad (16)$$

where the phonon creation operator $\tilde{\beta}_Q^\dagger$ includes the two processes with an overall coupling constant

$$|\tilde{C}(Q)|^2 = \int \frac{dq_f}{(2\pi)} |\alpha(Q, q_f)|^2 |[I_{2,2}(q_f) - I_{1,1}(q_f)]|^2 : \quad (17)$$

given the identity of the final states, the two processes are coherently summed in amplitude.

Inserting the explicit form of the Frölich coupling into this expression gives

$$|\tilde{C}(Q)|^2 = \frac{\pi \hbar \omega_{phon} e^2}{Q \epsilon_\rho} F(Q) \quad (18)$$

where the details of the quantum well are summarized in the form factor

$$F(Q) = \int dz \int dz' e^{-Q|z-z'|} [|\phi_1(z)|^2 - |\phi_2(z)|^2] [|\phi_1(z')|^2 - |\phi_2(z')|^2] . \quad (19)$$

For the typical processes under consideration, the phonon wavevector Q is much smaller than the well thickness L_w , so the form factor can be expanded at lowest order in Q . This leads to an approximately Q -independent coupling constant

$$\kappa^2 = \frac{\pi f \hbar \omega_{phon} e^2 L_w}{\epsilon_\rho} \quad (20)$$

where the numerical coefficient f is, in the most relevant case of a square quantum well with infinite barriers, $f \approx 0.032$. The constant value of κ allows for a straightforward reformulation of the Hamiltonian in real-space in the form

$$H^{ISB-ph} = \kappa \int d^2\mathbf{r} \Phi^\dagger(\mathbf{r}) \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}) + \text{H.c.} \quad (21)$$

where $\Phi(\mathbf{r})$ and $\Psi(\mathbf{r})$ are two-dimensional bosonic operators describing the phonon and the ISB excitation fields. These are coupled by a spatially local, nonlinear three-operator term describing parametric processes. For realistic values of the system parameters such as $L_w = 8.3$ nm, $\epsilon_\infty = 10.9$ and $\epsilon_s = 12.9$ (giving $\epsilon_\rho \sim 70$), one gets $\kappa \sim 30$ meV nm.

This interaction Hamiltonian is directly extended to an analogous coupling to ISB polaritons by including the Hopfield coefficient u_{ISB} quantifying the ISB weight of the polariton. Note that, depending on the specific configuration at hand, some care might be required in including the momentum-dependence of u_{ISB} . A straightforward Fermi golden rule calculation based on (21) provides a prediction

$$\Gamma_{sp} = \frac{S_k}{2\pi\hbar} \kappa^2 |u_{ISB}^{in}|^2 |u_{ISB}^{fin}|^2 \frac{2}{\pi\hbar\Gamma_{pol}} \quad (22)$$

for the spontaneous phonon emission rate by polaritons into an area S_k of final states in momentum space, corresponding to a solid angle $d^2\Omega \approx S_k(c/\omega_{pol})^2$ for collection of scattered light. Given

the relatively large value of the polariton linewidth Γ_{pol} , the weak dispersion of polariton states along the direction y orthogonal to the patterning gives a negligible contribution. Using the experimental parameters and a collection cone with a radius of 5° , one obtains a spontaneous scattering rate on the order of $5 \cdot 10^4 \text{ s}^{-1}$, that is a factor around 10^{-8} smaller than the overall polariton decay rate. This value is comparable to the one predicted and measured in [36] for a dispersive cavity.

As a final comment, it is interesting to note how this phonon relaxation rate is relatively slow compared to the phonon-electron relaxation time, typically in the ps range. This is mostly due to the much lighter mass of the polariton modes (in dispersive configurations) or to the combination of a large polariton decay rate γ_{pol} and the small momentum-space area of the relevant collected polaritons (in non-dispersive configurations).

4.1.2. Generalized optical Bloch equations

In order to obtain quantitative predictions in view of experiments for stimulated phonon-polariton scattering processes [32], we can include the ISB-phonon coupling term (21) into the formalism developed in [37] for the description of the full nonlinear dynamics of ISB polaritons. Beyond the bosonic description, this also includes saturation and intensity-dependent shift of the ISB transition [10,16,17,18,38,60–65].

This formalism is based on coupled differential equations for the cavity-photon field $E(\mathbf{r}, t)$, the ISB coherence $\sigma(\mathbf{r}, t)$, and the ISB population difference $\Pi(\mathbf{r}, t)$, plus the additional classical field $\Phi(\mathbf{r}, t)$ describing the phonons,

$$i \partial_t E = \left(\omega_{cav} - i \frac{\Gamma_{cav}}{2} \right) E - \Omega_R \sigma + E_{ext}(\mathbf{r}, t) \quad (23)$$

$$i \partial_t \sigma = \omega_{ISB} \sigma + \Omega_R \Pi E + \eta(\Pi + 1) \sigma - i \frac{\Gamma_{coh}}{2} \sigma + \kappa(\Phi + \Phi^*) \sigma \quad (24)$$

$$\partial_t \Pi = 2i \Omega_R (E \sigma^* - E^* \sigma) - \Gamma_{pop}(\Pi + 1) \quad (25)$$

$$i \partial_t \Phi = \omega_{phon} \Phi + \kappa n_{el} |\sigma|^2 - i \frac{\Gamma_{phon}}{2} \Phi. \quad (26)$$

The bosonic ISB excitation field Ψ in (21) is related to the ISB coherence field σ in the OBEs by $\Psi = \sqrt{n_{el}} \sigma$ where n_{el} is the in-plane electronic density in the system. As compared to the bosonic model of (21), saturation of the ISB transition is included by the value of the population raising above the ground state value $\Pi = -1$: this provides a reduction of the Rabi frequency in (24). A population-dependent depolarization shift is also visible, as discussed in [39]

Here, $\omega_{cav, phon, ISB}$ are the frequencies of the cavity mode, of the LO phonon, and of the ISB excitation (including the depolarization shift). Depending on the specific cavity configuration, the cavity mode may display a marked dependence on the in-plane wavevector, that translates into suitable differential operators in real-space. Ω_R is the Rabi frequency of the coupling between light and the ISB transition. $E_{ext}(\mathbf{r}, t)$ is the (suitably normalized) amplitude of the coherent incident field driving the system. η is the depolarization shift at linear regime, $\Gamma_{phon, pop, coh}$ are respectively the decay rates of the phonon mode, of the population, and of the coherence.

In what follows we will focus on the case of spatially wide pump and probe beams, so we can project the partial differential Eqs. (23–26) into the few relevant optical modes at the pump $\mathbf{k}_p$

and probe $\mathbf{k}_s$ wavevectors, and on the corresponding phonon mode Φ_q at $\mathbf{q} = \mathbf{k}_p - \mathbf{k}_s$,

$$i \partial_t E_{s,p} = \left(\omega_{cav}^{s,p} - i \frac{\Gamma_{cav}}{2} \right) E_{s,p} - \Omega_R \sigma_{s,p} + E_{ext}^{s,p}(t) \quad (27)$$

$$i \partial_t \sigma_s = \omega_{ISB} \sigma_s + \Omega_R (\Pi_0 E_s + \Pi_q^* E_p) + \eta [(\Pi_0 + 1) \sigma_s + \Pi_q^* \sigma_p] - i \frac{\Gamma_{coh}}{2} \sigma_s + \kappa \Phi_q^* \sigma_p \quad (28)$$

$$i \partial_t \sigma_p = \omega_{ISB} \sigma_p + \Omega_R (\Pi_0 E_p + \Pi_q E_s) + \eta [(\Pi_0 + 1) \sigma_p + \Pi_q \sigma_s] - i \frac{\Gamma_{coh}}{2} \sigma_p + \kappa \Phi_q \sigma_s \quad (29)$$

$$\partial_t \Pi_0 = 2i \Omega_R (E_s^* \sigma_s + E_p^* \sigma_p - E_s \sigma_s^* - E_p \sigma_p^*) - \Gamma_{pop} (\Pi_0 + 1) \quad (30)$$

$$\partial_t \Pi_q = 2i \Omega_R (E_p \sigma_s^* + E_s^* \sigma_p) - \Gamma_{pop} \Pi_q \quad (31)$$

$$i \partial_t \Phi_q = \omega_{phon} \Phi_q + \kappa n_{el} \sigma_s^* \sigma_p - i \frac{\Gamma_{phon}}{2} \Phi_q \quad (32)$$

Here, $E_{p,s}$ and $\sigma_{p,s}$ are the spatial Fourier components of the electric field amplitude and of the electronic coherence at respectively the pump and probe wavevectors, while $\Pi_{0,q}$ are the spatial Fourier component of the population $\Pi(\mathbf{r})$ at respectively zero wavevector and at wavevector $\mathbf{q}$ (reality of $\Pi(\mathbf{r})$ imposes that $\Pi_{-q} = \Pi_q^*$). As in this work the parametric process involves the (stimulated) scattering of a polariton at $\mathbf{k}_p$ into a polariton at $\mathbf{k}_s$ plus a phonon at $\mathbf{q}$, there is no third polariton mode involved in the process as it was instead the case in [37].

4.2. Analytical estimates

Before moving to the numerical analysis, we gain analytical insight into the feasibility of stimulated phonon scattering by estimating the parametric oscillation threshold. We thus consider the pump-only steady state and evaluate its stability against growth of probe polariton and phonon components. In the pump-only steady-state, one has that $\sigma_p(t) = \sigma_p^{(ss)} e^{-i\omega_p t}$ oscillates at ω_p , while $E_s, \sigma_s, \Pi_q, \Phi_q$ instead vanish. For sufficiently large Ω_R in the strong-coupling regime, the pair of equations for E_s and σ_s can be combined giving independent equations for the two polariton modes at wavevector $\mathbf{k}_s$.

Focusing on the equation for the field ψ_{pol}^{fin} in the final lower-polariton state, linearizing it around the zero-field pump-only steady-state, and combining with the analogously linearized equation for the phonon field Φ_q , one gets a pair of linear equations in the form

$$i \partial_t \Phi_q = \omega_{phon} \Phi_q - i \frac{\Gamma_{phon}}{2} \Phi_q + \kappa n_{el} u_{ISB}^{fin} \sigma_p \psi_{pol}^{fin*} \quad (33)$$

$$i \partial_t \psi_{pol}^{fin} = \omega_{pol}^{fin} \psi_{pol}^{fin} - i \frac{\Gamma_{pol}^{fin}}{2} \psi_{pol}^{fin} + \kappa u_{ISB}^{fin} \sigma_p \Phi_q^* \quad (34)$$

where ω_{pol}^{fin} and $\Gamma_{pol}^{fin} = |u_{ISB}^{fin}|^2 \Gamma_{coh} + |u_{cav}^{fin}|^2 \Gamma_{cav}$ are respectively the frequency and the decay rate of the final ISB polariton state, this latter resulting from an average of the cavity and ISB transition linewidths $\Gamma_{cav,coh}$ weighted by the Hopfield coefficients $u_{ISB(cav)}^{fin(in)}$ of the final (initial) polariton state on the ISB transition (cavity mode). From the dynamical stability of these linearized equations in the resonant $\omega_{phon} + \omega_{pol}^{fin} = \omega_p$ case, one can derive the threshold for parametric oscillation

$$\kappa^2 n_{el} |u_{ISB}^{fin}|^2 |\sigma_p^{(ss)}|^2 = \frac{\hbar^2 \Gamma_{pol}^{fin} \Gamma_{phon}}{4}. \quad (35)$$

In our case, the condition $|\sigma_p|^2 \leq 1$ holding for any saturable emitter provides a general bound

$$\kappa^2 n_{el} |u_{ISB}^{fin}|^2 > \frac{\hbar^2 \Gamma_{pol} \Gamma_{phon}}{4} \quad (36)$$

to observe a significant gain before the ISB transition gets saturated. Eq. (36) suggests that higher electronic densities are favorable in making the ISB transition robust against saturation.

This theory can be generalized to the case of many quantum wells coupled to the cavity: the reduction by a factor $\sqrt{N_{QW}}$ of the phonon-polariton coupling κ due the delocalization of both the phonon and the ISB polaritons among the different wells is compensated by the increased total number of electrons (at fixed electron density per well n_{el}) which conspire in maintaining the same relation (36) for any number of wells. Of course, having a larger N_{QW} is favourable in view of increasing the Rabi splitting between the ISB polariton branches. Finally, it is worth highlighting how the small decay of phonons offers important advantages compared to the purely polariton scattering processes of [37]: on one hand, a small Γ_{phon} facilitates reaching the condition (36) for gain; on the other hand, the reduced linewidth of the parametric gain peak gives an easier recognizable feature in the probe spectra as shown in the next Subsection.

Using this same argument, we can obtain an estimate for the incident pump power that is needed to reach the lasing threshold. Under a weak saturation condition $|\sigma_p|, \Pi + 1 \ll 1$, we can assume a linear response of the ISB transition to incident light. Around critical coupling to the incident radiation, most of the incident power enters the cavity. It is then possible to relate the in-cavity ISB polariton density to the incident power via

$$I_{inc} = \eta \hbar \omega_{pol}^{in} \Gamma_{pol}^{in} \bar{n}_{in} \quad (37)$$

where $\bar{n}_{in}$ is the polariton density in the initial mode and η is a coefficient of order 1 accounting for the details of the radiative process. Combining (37) with the general condition for parametric oscillation (35) and keeping in mind that the electronic coherence is related to the polariton density via $n_{el} |\sigma_p|^2 \sim \bar{n}_{in} |u_{ISB}^{in}|^2$, we get to the final estimate for the threshold intensity

$$I_{th} = \frac{\hbar^2 \Gamma_{phon} \Gamma_{pol}^{fin} \Gamma_{pol}^{in} \omega_{pol}^{in} \epsilon_p}{4\pi \omega_{LO} e^2 f L_{QW} |u_{ISB}^{in} u_{ISB}^{fin}|^2} \quad (38)$$

Using realistic parameters for our experimental device, we obtain a rather optimistic value in the 50 kW/cm² range.

4.3. Results of numerical simulations

Our numerical simulations are based on a set of experimental parameters. The complete polaritonic band is implemented in the model to reflect the non-dispersive nature of the system. This band dispersion is obtained from Rigorous Coupled Wave Analysis simulations, which are fitted to experimental data. The ISB transition frequency (ω_{ISB}) is set to 27.3 THz, with a linewidth (Γ_{coh}) of 1.6 THz, consistent with state-of-the-art experimental values. The Rabi frequency ($\Omega_R = 3$ THz) and the cavity linewidth ($Q \approx 20$) are chosen to match the observed experimental behavior. The system is modeled at the anti-crossing point, where the two polaritonic modes form perfectly mixed light-matter states. For the LO_{ph} in GaAs, we use parameters from [66]: a central frequency (ω_{phon}) of 8.75 THz, a damping rate (Γ_{phon}) of 69.8 GHz. For the effective coupling constant $\kappa \sqrt{n_{el}}$, we calculate from (20) a value of 0.81 THz using the electron density $n_{el} = 1.4 \times 10^{12}$ cm⁻² taken from the experimental Rabi splitting. The nonlinear coefficient in the OBE ($\eta = 5$ THz) is set in agreement with the value reported in [37].

To probe the final state and pump the ISB polariton reservoir, we define two optical pulses: a probe pulse modeled as an ultrashort Gaussian pulse with a duration of 70 fs, centered at 27 THz; and a pump pulse modeled as a quasi-continuous wave (quasi-CW) pulse centered on the upper polaritonic mode at 31 THz, with its duration optimized to enhance the stimulated scattering process. Due to the narrow linewidth of the LO_{ph} and the relatively low coupling efficiency, our numerical simulations indicate that the system reaches the stimulated regime only when using long pump pulses on the order of hundred of picoseconds.

Figure 5(a) shows the simulated reflectance of the polaritonic system both without the pump and with the optimal pump pulse parameters (power and duration, $\tau_{pump} = 200$ ps) chosen

to clearly reveal amplification. As expected, the amplified signal appears exactly 8.75 THz below the pump frequency, and the absolute reflectance reaches a value three times greater than unity, indicating the presence of optical gain. To further analyze this effect, Fig. 5(b) presents the temporal evolution of the probe field before and after interacting with the sample under illumination, together with the generated coherent phonon field. The inset of Fig. 5(b) shows in detail how the probe field is modified due to the stimulated scattering of the polariton population into the final state. Notably, there is a clear correspondence between the evolution of the probe field and the emergence of the coherent phonon field, as highlighted by the temporal zoom shown in the inset.

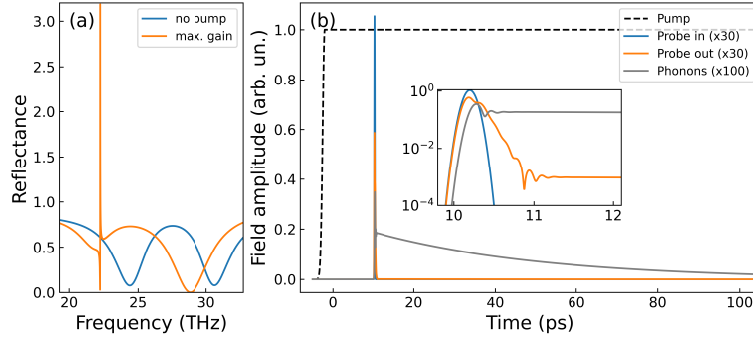


Fig. 5. (a) Calculated reflectance of the sample at the conditions of no pump and of maximum gain. (b) The absolute value of the pump field (dashed line), the probe field sent to probe the system (blue), the probe field after probing the system (orange), along with the phonon field (grey). The inset shows a zoom of the fields in logarithmic scale.

As shown in [37], amplification is closely tied to polariton saturation: as the ISB reservoir is pumped, the population difference—and thus Ω_R —decrease, renormalizing the system. Consequently, we perform a simulation in which the pump pulse energy (E_p) is gradually increased. Figure 6(a) presents a colormap of the sample reflectance as a function of pump pulse energy. At low pulse energy, both polaritonic states are visible, indicating that the system remains almost unperturbed. As the pump pulse energy increases, we observe a decrease in Ω_R , accompanied by a strong redshift of the upper branch and a less pronounced one of the lower one. Interestingly, a sharp feature appears one LO_{ph} below the pump frequency, at 22.25 THz, which is the optical gain feature already shown in Fig. 5(a). Figure 6(b) shows a cut of the sample reflectance at this specific frequency, where a net optical gain is observed, with the reflectance exceeding unity (area shaded in blue). To illustrate the close relation with saturation, Fig. 6(c) shows the population difference between the two subbands as a function of pump pulse energy. The dashed line marks the pump energy at which saturation is reached (i.e., $\Delta n = 0.5$), while transparency is reached at $\Delta n = 1$. The rapid gain decline results from the combined reduction in Ω_R and redshift of the injection state, which lowers the coupling efficiency into the latter.

While our OBE framework uses normalized units, we can estimate the saturation value indicated in Fig. 6(b,c) using temporal coupled mode theory [19], which yields a saturation intensity around 130 kW/cm², in agreement also with the prediction from (37). Noting from Fig. 6(b) that the ratio of the threshold pump energy and the saturation pump energy is around 2/3, this value is consistent with the threshold intensity of 50 kW/cm² derived in Section 4.2. Such levels can be readily achieved using a DFG system pumped by a multi-pass OPA stage. The pulse duration can range from several tens of picoseconds up to nanoseconds. In the nanosecond regime, pumping with solid-state sources such as QCLs could also be considered, allowing intensities up to 100 kW/cm².

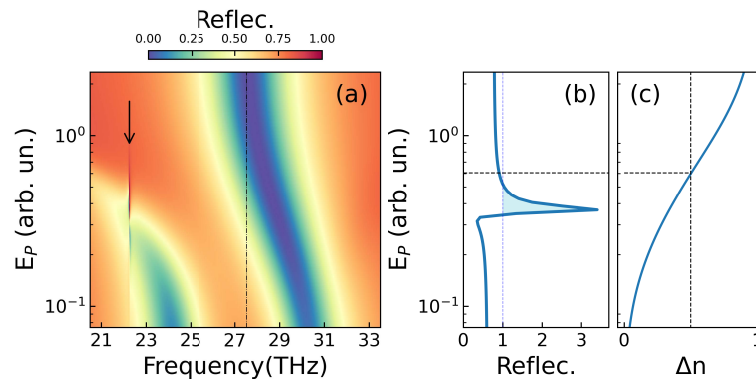


Fig. 6. (a) Simulated Reflectance of the probe pulse as a function of the energy of the pump pulse (E_P) and of the probe frequency. The arrow indicates the frequency at which the gain ($R > 1$) appears on the probe. The dash line marks the bare cavity resonance. (b) Reflectance of the probe pulse at a frequency of 22.25 THz as a function of the energy of the pump pulse. The blue colored area below the curve shows the region where one can observe gain. (c) Population difference between the two subbands of the QW as function of the energy of the pump pulse. Dash lines indicates the value at which the transition is saturated, having a quarter of the population in the 2nd subband.

5. Conclusion

In summary, this work has experimentally demonstrated the interaction between ISB polaritons and LO phonons in non-dispersive MIM cavities. The spontaneous emission of polaritons that scattered with LO-phonons from the upper branch is weak, primarily due to the combination of a large polariton decay rate (γ_{pol}) and the limited momentum-space area available for the collected polaritons. We have then numerically shown that implementing a pump–probe scheme could enable the system to reach the regime of final-state stimulation, where optical gain can build up on the probe pulse. Although this result is exploratory, the predicted conditions for observing stimulation and gain appear experimentally achievable, paving the way for the realization of inversionless lasing and potentially polariton condensation at MIR and FIR wavelengths.

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